Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Modular customizable active elbow exoskeleton*
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ), Others (x)

Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical			
	Medium Category	Small Category	Medium Category	Small Category
	Functional Restoration / Assistive & Welfare Devices	• *Physical Function Restoration Devices* • *Rehabilitation Training Devices* • *Mobility Support Devices* • *Daily Living Support Devices and Systems* • *Other Functional Restoration / Assistive &Welfare Devices*		

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Concept of Proposed Technology	*Core concept:* *We propose a modular, lightweight external assistive system to compensate the brachioradialis during elbow rehabilitation, using flexible drive trains (bendable motors, Bowden cables, Twisted String Actuators) and EMG-driven control for intuitive operation. The main functions include delivering targeted assistive torque to support elbow flexion/extension and pro-/supination synergistic to the M. brachioradialis, which stabilizes the forearm, aids rotation, and enables precise movements [2, 6, 9]. The system features adjustable lever lengths and attachment points to fit individual anatomy and rehab phase, variable support levels, easy assembly/disassembly, and a slim, low-profile design. Power efficiency is achieved through EMG-controlled actuation with AI intention detection, with a total system weight under approximately 500 g (including the battery).* *Potential uses span rehabilitation for radial nerve injury [3], brachial plexus injuries [8, 13], cervical spinal cord injuries [10], stroke [17], compartment syndrome [7], tennis elbow [11], or rheumatoid arthritis [5]. The device is supposed to be suitable for home-based therapy and clinical rehabilitation in hospitals and can enhance daily living activities and independence, thereby improving quality of life, and serves as a research platform for biomechanics, control strategies, and EMG interfaces.*
Rationale for Joint R&D Project	*There is a clear gap in current elbow rehabilitation devices, which are often bulky, poorly adjustable to fit individual anatomy and rehabilitation progression, and offer limited biomechanical realism because they mainly compensate elbow flexion [14, 16]. This gap is underscored by demographic demand, with aging populations and rising incidences of stroke, spinal cord injuries, nerve injuries, and arthritis, alongside the growing need for home-based rehabilitation.* *Collaboration is required because progress demands multidisciplinary integration across biomechanics, robotics and mechatronics, neuromuscular control, rehabilitation science, materials, and user experience. Clinical access to hospitals and rehabilitation centers is essential for real-world testing and validation. Industrial expertise and manufacturing technologies are also necessary. Regulatory translation is needed to address safety, ethics, trial design, and approvals.*
Key Words	*1. Active Exoskeleton* *2. Elbow support* *3. Rehabilitation technology* *4. Usability* *5. Adaptability*

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 1	End of the Project	Level 4

R&D Objectives	*1. Conceptualization and Architecture Development, including Biomechanical Modeling*
	• *Architecture for adjusting elbow movement angles and torques (flexion/extension and pro-/supination) to be adaptable per patient and rehab phase*
	• *Interchangeable lever arms for flexion/extension and pro-/supination via multiple attachment points above and below the elbow around the upper and lower arm*
	• *Flexible drive-train concept: wearable-friendly routing/positioning around the body, comfortable during movement*
	• *Biomechanical modelling: digital twin development, simulations to investigate biomechanical effects of the exoskeleton on musculoskeletal strain*
	
	2. Drive Train and Control Engineering
	• *Actuators and drive-train investigations for flexible, lightweight, adjustable support (modularity of actuators)*
	• *Investigation of various concepts:*
	  ○ *Flexible linear motors*
	  ○ *Twisted String Actuators (TSA): lightweight, compact linear actuators for displacement along curved paths*
	  ○ *Cable-driven system vs standard electric motor configurations*
	• *EMG-driven control system: real-time intent detection and proportional assistance with safety limits (AI-driven)*
	• *Integrated sensor suite for closed-loop control: EMG, elbow angle (encoder/IMU), and force/torque sensing*
	• *Embedded control platform: low-latency edge computing, robust safety features, fault handling*
	• *Lightweight power system: battery and power management to keep total device weight under ~500 g*
	
	3. Interface Engineering
	• *Adjustable ergonomics: modular harness and attachment interfaces for individual anatomy and rehab phase*
	• *Sensor/interface integration: data fusion and user feedback for EMG, angle, and torque sensing*
	
	4. Clinical Validation
	• *Regulatory and safety framework outputs: risk assessment and documentation*
	• *Safety case and compliance: hazard analysis, risk mitigation, and fault-handling strategies*			

R&D Target Specifications	• *Generated force: 5–15N (Required to compensate Brachioradialis during flexion/extension) [1]* • *Elongation/shortening: approx. 20% [1]* • *Shortening speed: e.g., 5 cm with a muscle length of 25 cm over 3–4 seconds* • *Range of motion support: 0–180° flexion/extension* • *Active assistance across essential daily tasks* • *System weight: ≤ 500 g total (including battery)* • *Battery life: ~2 hours continuous assist; standby 8–12 hours* • *EMG sampling: Real-time, responsive control,* • *Latency (EMG-to-actuation): < 40 ms*			

	• *Sensing accuracy: Angle: ±0.5° and Force/Torque: ±5%* • *Endurance 100k actuation cycles* • *Usable in typical rehab settings and at home* • *Donning/adjustment time: ≤ 1 minute* *→ The aim is to develop and evaluate three concepts with different actuation systems.*

R&D Plan	*The research approach will be iterative and user-centered, guided by risk-based testing to inform design decisions. Our work will require multidisciplinary integration across biomechanics, mechatronics, neuromuscular control, rehabilitation science, wearables, and user experience. Hardware-in-the-loop testing and biomechanical validation will be performed early and throughout the development process. A clear regulatory and safety plan will be established from the outset to address compliance, risk management, and trial design.*
	
	Work packages:
	• *WP1: Requirements, use cases, and risk management (months 0–3)*
	  ○ *Tasks: stakeholder definition and interviews, clinical need assessment, safety/risk plan, use-case scenarios.*
	  ○ *Methods: surveys, expert workshops, risk analysis plan*
	• *WP2: Conceptual design and modelling (months 1–4)*
	  ○ *Tasks: architectural concepts (modular architecture, harness, actuators), biomechanical modelling (multi-body elbow model), simulation of ROM and torque.*
	  ○ *Methods: Design-thinking, CAD sketches, finite element/ multibody simulations.*
	• *WP3: Actuators and drive technologies (months 3–12)*
	  ○ *Tasks: develop prototypes with different drive train and actuation technology*
	  ○ *Methods: rapid prototyping, material testing, load–deflection tests, efficiency measurements.*
	• *WP4: EMG-Sensing and control (months 6–14)*
	  ○ *Tasks: EMG-driven control algorithms, sensor suite integration (EMG, angle/torque sensing), safety features (faults, overcurrent, thermal).*
	  ○ *Methods: real-time control algorithm development, simulations, hardware-in-the-loop testing.*
	• *WP5: User interface (months 3–10)*
	  ○ *Tasks: develop adaptable and comfortable interfaces and operation elements, ensure adaptability*
	  ○ *Methods: textile engineering, interface engineering*
	• *WP6: System integration and verification of the functionality (months 12–22)*
	  ○ *Tasks: integrate modules into a cohesive prototype, create test benches and experimentally verify functionality and basic usability.*
	  ○ *Methods: hardware integration, testing in test rigs, usability testing, design iterations and optimization*
	• *WP7: Clinical validation (months 18–30)*
	  ○ *Tasks: create documentation and other regulatory preparations, define subject study methodology for wearability and comfort assessments, usability metrics and functionality in clinical context*
	  ○ *Methods: study design, subject tests with user surveys and analysis.*
	
	Milestones:
	Phase I: Creating and validating a biomechanically sound concept
	• *M1: Requirements and risk plan approved (month 3)*
	• *M2: Concept defined (month 4)*
	• *M3: Biomechanical model validated (month 6)*
	Phase II: Developing and clinically validating the feasibility of the approach

	• *M4: Alpha actuator prototypes validated in bench tests (month 12)* • *M5: EMG control framework validated in bench tests (month 14)* • *M6: Integrated prototype (months 22)* • *M7: Feasibility study and regulatory preparations complete (month 30)* *Risks and mitigations:* *There is a risk of mechanical mismatch with the anatomy, which will be mitigated by using modular lever arms, an adjustable harness, and early biomechanical validation. EMG control variability is a risk, but mitigations include robust signal processing, user calibration, and redundancy in sensing. Regulatory delays are possible. Mitigations involve parallel safety case development and early liaison with regulators. Integration complexity could hinder progress. Mitigations include a modular architecture, well-defined interface standards, and frequent integration tests.*

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*The Biomechatronic Systems Department at the Fraunhofer Institute for Manufacturing Engineering and Automation (IPA) has extensive experience in biosignal acquisition and in developing high-level control strategies for upper-extremity exoskeletons, and is a leader in the development and control of active upper-extremity exoskeletons. This particularly concerns mechanical and ergonomic development, including comfort and usability engineering, the recording of biosignals, and the development of control strategies for exoskeletons.* *Proposed allocation of responsibilities in the collaboration framework: We will develop architectural concepts, validate biomechanical efficacy using biomechanical models (WP2), and design and build integrated prototypes (WP6). Hence, we will draw on our broad experience in the field and in most of the topics to integrate actuators, sensing technologies, and comfortable user interfaces (WP3–5), and to develop the high-level control of the final systems (WP4), while others take the lead in the aforementioned development tasks for individual components (actuators (WP3), interfaces (WP5)), requirement specifications (WP1) and clinical validation (WP7).*
Expected R&R of Korean SME Partner	*Potential project partners have synergistic competencies in interface design, actuation, or product development, as well as in the testing and certification of medical devices. We are specifically looking for OEMs in medical devices or orthopedics, as well as producers/developers of specialized actuator technologies that fit the approach (for example, but not limited to flexible linear motors, twisted string actuators, cable-driven actuators, pneumatic muscles, and miniature electric motors).* *Medical and orthotic device companies' responsibilities could include defining a clinical use case and strategy for the minimum viable product, gathering and defining user requirements (WP1), engineering interfaces in detail (WP5), and managing the documentation of the future product (risk assessment, regulatory strategy and compliance, quality management), as well as preparing and conducting legally and ethically compliant end-user tests (WP7).* *Actuation company responsibilities could include use-case-specific actuator technology development, miniaturization, implementation of low-level control and safety control, and testing of the final actuators (functionality, performance characterization, reliability) (WP3). They would also investigate and account for regulatory and economic aspects of their drive units.*

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	*The modular elbow-support architecture enables interchangeable actuator modules and adjustable levers that adapt to individual rehab phases and capabilities.* *The aim is to develop three prototypes with different actuation principles and an EMG-driven control strategy. Biomechanical validation and user-centered design will help us gain knowledge about the biomechanics of the elbow and its translation into a technical assistive device.* *The project envisions commercialization potential in laying the foundation for market-ready solutions for home and clinical rehabilitation, driven by a scalable modular architecture and an IP strategy aimed at patents on the modular framework, drive train concepts, and the EMG-based control approach. Regulatory pathways can be mapped for the EU, Korea, and other regions, with concrete plans for clinical trials and approvals, and there are established partnerships with clinics, rehab centers, and reimbursement organizations to support adoption and reimbursement. The broader industrial and economic impact includes strengthening Korean SME capability in orthopedic robotics and actuators while fostering closer collaboration with European partners, as well as creating jobs in design, prototyping, testing, and regulatory affairs and enhancing both domestic and international supply chains. The initiative will contribute to the advancement of standards for EMG-driven orthotics and modular rehabilitation devices and holds potential for healthcare cost savings through home-based rehabilitation and reduced therapy time, ultimately improving patient independence and quality of life.*

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization